Gennoor Tech

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GENNOOR ACADEMY · SELF-PACED · 35 MIN

Demand Forecasting in Practice

A ~35-minute course for CSCOs, S&OP heads, demand planners — landscape, accuracy and the MAPE trap, model choice, demand sensing, promo uplifts, NPI, S&OP integration, 2-quarter rollout.

8	35 min	Intermediate	FREE
CHAPTERS	DURATION	LEVEL	CERTIFICATE

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Course Overview

For CSCOs, S&OP heads, demand planners, operations directors past basic AI literacy. Eight chapters across the demand-forecasting surface: the forecasting landscape (Gartner 78% adoption / 41% measurable lift, McKinsey 15–30% MAPE reduction / top-decile 30–50%, M5 LightGBM domination, Hyndman canonical reference, Tetlock calibration > confidence); accuracy and the MAPE trap (APICS +5–12% structural bias, MAPE asymmetry, sMAPE/MASE/RMSSE alternatives, M5 used RMSSE, APICS $0.5–1.5M per percentage point); choosing the right model (3-variable decision tree — history × signal × granularity, LightGBM granular, classical aggregated, Croston intermittent, ensemble options); demand sensing (Blue Yonder + o9, POS/ weather/social signals, 5–15% lift on 4-week horizon, confidence-weighted overrides, noise-chasing risk); promotional uplifts (Coca-Cola/Unilever/Carrefour 20–40% improvement, 3-component pattern, calendar drift failure mode); NPI (P&G McKinsey case 2024, analog-product lookup + first 4-week signal, judgement layer); S&OP integration (SCC 38% gap, FVA analysis, LokadTech LLM hallucination case, quote-or-cut narrative); and the close — 2-quarter rollout playbook with interactive builder, 4 trust trip-wires, through-line: planners with ML beat planners without it, planners plus ML still beats ML alone, calibration matters more than confidence.

FORMAT: Self-paced · 8 chapters · audio-narrated slides · interactive end-of-course builder · free certificate on completion

Who Is This For?

1. **CSCOs · S&OP heads**
2. **Demand planners**
3. **Supply chain analysts**
4. **Operations finance**

Learning Outcomes

1. The honest landscape — M5 Forecasting Competition reality, where AI beats classical and where it doesn't
2. Accuracy and the MAPE trap — why FVA (Forecast Value Add) is the metric that matters
3. Model choice — when classical wins, when ML wins, when LLM-augmented wins
4. Demand sensing in practice — Coca-Cola / Unilever / Carrefour cases, 20-40% promo-week lift
5. Promotional + event uplifts — clean promo calendar + LightGBM uplift model + cannibalisation
6. New product introduction — P&G analog-product approach, ML does the math, judgement picks the analogs
7. S&OP integration — narrative + accountability, FVA analysis, the quote-or-cut rule against LLM hallucination
8. A 2-quarter rollout with 4 trust trip-wires — no unexplainable model, no unrecorded override, no unsourced AI narrative, no set-and-forget

8-Chapter Curriculum

Self-paced, audio-narrated slide format. Each chapter ends with a "Monday move" the learner can apply that week.

01 The forecasting landscape

CHAPTER 1 · 5 MIN · NARRATED + INTERACTIVE

- Gartner SCT25 2025: 78% have AI-in-forecasting initiative, 41% measurable lift.
- McKinsey: 15–30% MAPE reduction (top quartile), 30–50% top decile.
- M5 LightGBM domination, classical competitive on aggregates.
- SAP IBP/Oracle/Blue Yonder/o9/Kinaxis.
- Tetlock: calibration > confidence.

02 Accuracy, bias, and the MAPE trap

CHAPTER 2 · 5 MIN · NARRATED + INTERACTIVE

- APICS 2024-25: median bias +5 to +12% (over-forecasting, structural).
- MAPE asymmetry — biased toward under-forecasting, catastrophic on intermittent demand.
- M5 used RMSSE.
- Use sMAPE/MASE/RMSSE by context.
- APICS: 1pp MAPE = $0.5–1.5M per $100M.

03 Choosing the right model

CHAPTER 3 · 4 MIN · NARRATED + INTERACTIVE

- The 3-variable decision tree: history × signal × granularity.
- LightGBM for long-history granular, ETS/ARIMA/Theta for aggregated, Croston (or Croston-LightGBM hybrid, Boylan 2024) for intermittent, analog-lookup for short history.
- Ensemble for robustness.
- ML weak past 6 months.

04 Demand sensing

CHAPTER 4 · 4 MIN · NARRATED + INTERACTIVE

- Blue Yonder + o9.
- POS, weather, ad spend, social signals. 5–15% MAPE reduction on 4-week horizon.
- Confidence-weighted overrides + cap on magnitude.
- Signal hygiene (POS latency, mapping) + FVA shadow-mode before production.
- Noise-chasing failure mode.

05 Promotional & event uplifts

CHAPTER 5 · 4 MIN · NARRATED + INTERACTIVE

- Coca-Cola / Unilever / Carrefour 2024-25 cases: 20–40% accuracy improvement on promotional weeks.
- Pattern: clean promo calendar + LightGBM uplift model + explicit cannibalisation modelling.
- Calendar drift (planned "` executed) is the failure mode.

06 New product introduction

CHAPTER 6 · 4 MIN · NARRATED + INTERACTIVE

- P&G NPI case via McKinsey 2024.
- Analog-product lookup + first-4-week early signal.
- Category expert picks analogs (judgement layer), ML does the math.
- Lock forecast for first 4 weeks, then re-tune. 3 buckets: analog-rich, analog-thin, genuinely novel.

07 Connecting forecast to S&OP

CHAPTER 7 · 5 MIN · NARRATED + INTERACTIVE

- SCC 2025: only 38% connect ML forecast to executive S&OP decisions.
- The gap is narrative + accountability.
- LokadTech 2025 LLM hallucination case (phantom promotion).
- FVA analysis (Gilliland/SAS) measures where in the chain accuracy is added or lost.
- Quote-or-cut for S&OP commentary.

08 Making it stick: your demand-forecasting roadmap

CHAPTER 8 · 4 MIN · NARRATED + INTERACTIVE

- 2 use cases
- 2 quarters
- 1 FVA discipline. Q1 = baseline ML on top 100 SKUs + FVA. Q2 = demand sensing OR promo uplift. 4 trust trip-wires: no unexplainable model, no unrecorded override, no AI narrative without source check (LokadTech rule), no "set and forget". Interactive roadmap builder.